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Experimental study on bond durability of glass fiber reinforced polymer bars in concrete exposed to harsh environmental agents: Freeze-thaw cycles and alkaline-saline solution

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ABSTRACT

This study presents an experimental investigation on the bond durability of GFRP bars in concrete when subjected to harsh environments. The pullout specimens having different concrete covers were designed based on a created database to demonstrate the generality of the current experimental program. The freeze-thaw (FT) cycles, alkaline-saline (AS) solution, and both coupled effects were used to simulate environmental conditions in cold regions. The durability performance in terms of the failure mode, weight loss, relative dynamic modulus of elasticity, durability factor, as well as the bond strength, were measured and investigated accordingly. The test results revealed that the concrete cover with three times of the bar diameter was not sufficient to resist the environmental agents when exposed to weathering, including FT cycles, in which all the pullout specimens failed by concrete splitting. The coupled scenario of FT cycles and AS solution was observed to be the worst case among all the environmental conditions. Moreover, the analytical models: modified Bertero-Eligehausen-Popov (mBPE) model and Cosenza-Manfredi-Realfonzo (CMR) model, were calibrated by considering the environmental influences based on the experimental data to better demonstrate the degradation of GFRP-concrete bond.

1. Introduction

Conventional reinforced concrete (RC) structures subjected to aggressive environments are prone to corrosion-induced damages. Following the reduction of cross sectional area of reinforcing steel bar [1], the build-up of corrosion product, such as the ever-growing and expanding rust, results in cracking or even spalling of concrete. As a result, the damages degrade the bar-concrete interfacial bond, and thus have adverse impacts on the load-carrying capacity and long-term durability of RC structures. The reinforcement corrosion is the leading causes of malfunction or even failures of RC structures in the United States, Canada and most European countries [2–4]. To overcome this drawback, glass fiber-reinforced polymer (GFRP) composites have been recognized due to their superior corrosion resistance as alternative reinforcing bars to be implemented in the RC structures [5–8].

Wide acceptance and application of GFRP bars in construction industry requires comprehensive investigation of their structural behavior [9]. Bond development is of great importance to ultimately achieve the force transfer mechanism at the bar-concrete interface, thereby dominating the integrity and ductility of the structural member. Due to different material and mechanical behaviors, the bond mechanism of FRP bar-concrete is more complicated and quite different from that of steel reinforcement. Critical factors and their impacts on failure mode and bond strength of GFRP bars to concrete, have been extensively investigated in Refs. [10–15] and summarized in Ref. [1]. Besides, bond durability under harsh environmental agents, such as alkaline or saline solutions, elevated temperatures, and freeze-thaw (FT) cycles, also plays a crucial role in the long-term performance of the concrete structures reinforced with GFRP bars.

Bond degradation mechanism is a complex process that normally initiates from the bar surface. The two constituents of GFRP bars, glass fibers and resin matrix, tend to deteriorate in strength when exposed to wet alkaline environments. Considering that concrete presents highly alkaline with a pH of about 12.5–13.5, this may not only deteriorate the resin matrix due to hydrolysis of the ester group and hydroxide ions, but also damage the glass fibers

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due to leaching and etching [16,17]. Thus, it requires that the resin should be fully cured to provide appropriate protection to fibers. On the other hand, moisture could diffuse through the resin up to the fiber-matrix interphase or even to the fibers, resulting in hydrolysis and plasticization of the resin, as well as a bond loss of the fiber-matrix interphase. As the major load-carrying component of GFRP composites, glass fibers may even deteriorate in properties due to the moisture extracting ions from the fibers [18,19]. Moreover, bond degradation of GFRP bars to the surrounding concrete become more serious in the presence of moisture containing chloride ions [20,21]. Such phenomenon is inevitably encountered for structures in cold regions, where the concrete pore solution may be contaminated with chloride ions as normally found in de-icing salts [22]. In addition, FT cycles, another environmental agent, also need to be considered for those structures [23–26]. Damage in concrete due to FT cycles depends on the saturation of concrete. The water can permeate through the voids (or new developed microcracks) in concrete matrix, while the growth of ice crystals during repeated freezing process generates high pressure to concrete, thus leading to microcracks in concrete. As a result, this will break down the bar-concrete interface. Also, presence of concrete cracks also yields a low confinement to GFRP bar and thus causes less bond strength.

Therefore, this study aims to investigate the bond durability of GFRP bars to concrete, especially in cold regions. In order to simulate and approximate field conditions that the structures normally experience in service-life stage, GFRP bars embedded in concrete were designed and exposed to different weathering: a) FT cycles, b) alkaline solution contaminated with chloride ions were considered; and (c) elevated temperature to accelerate the degradation rate. Some critical indices associated with the environmental conditioning, such as the weight loss, the relative dynamic modulus of elasticity and durability factor, and the bond strength reduction, were measured to evaluate the durability performance of the GFRP-concrete elements. Moreover, the analytical models were integrated with experimental data to better demonstrate the degradation of GFRP-concrete bond.

2. Background and research to date

2.1. Bond durability

It is well established that aqueous solutions with high pH can reduce the tensile strength of bare GFRP bars despite test results showed great differences in Refs. [27–31]. Also, considerable studies have gone into various environmental attacks on the bond strength of GFRP bars. However, the combined effect of the environmental agents including alkaline solution, saline solution, and freeze thaw cycles, remains unsolved or even holds contrary opinions.

Chen et al. [16] used several types of solutions, the tap water, alkaline solutions with respective pH of 12.7 and 13.6, saline solution, and alkaline solution contaminated with chloride ions, to investigate the durability of bare GFRP bars and GFRP-concrete elements. Those specimens also experienced FT cycles and wet-dry cycles before testing. Significant reductions in tensile strength and bond strength were observed for the respective bare and embedded GFRP bars. Alkali attack was stated to be more serious than the FT cycles and wet-dry cycles. Davalos et al. [9] reported bond performance of GFRP bars in concrete subjected to different environmental conditions: tap water at normal temperature and 60 °C, thermal cycles ranging from 20 to 60 °C. They reported that there were 0–20% reductions in bond strength being observed for the GFRP bars. Similar results of the bond strength reduction can also be found in Ref. [32]. Fursa et al. [33] conducted experiments on sixteen GFRP-concrete samples subjected to FT cycles. They used the electric response to evaluate the bond strength, and found that the bond strength reduced nearly 50% after 18 FT cycles ranging from −40 to 20 °C.

On the contrary, Mufti et al. [34,35] conducted studies on five field GFRP reinforced concrete bridge structures exposed to natural environments for durations of five to eight years. The environmental conditions encompassed FT cycles, wet-dry cycles, de-icing salts, thermal range from −35 to 35 °C. The GFRP bars in those selected demonstration structures were all composed of E-glass and vinyl ester resin. The analysis results stated that the structures maintained a good bond at the GFRP bar-concrete interface and no degradation was observed by either optical microscope or Fourier transformed infrared spectroscopy. Robert and Benmokrane [36] performed experimental investigation on the bond durability of GFRP bars embedded in concrete. The specimens were exposed to tap water at different temperatures (23, 40, 50 °C) for three immersion durations (60, 120, 180 days). It was concluded that the bond strength decreased as the exposure durations increased whereas minor reductions of the bond strength were observed with increasing the exposure temperature. They also conducted experiments of mortar-wrapped GFRP bar specimens immersed in saline solutions with 50 °C for 365 days and 70 °C for 120 days [7]. The micrographs showed that no significant damage was captured at the bar-concrete interface. Moreover, the bar-concrete interface and fiber-matrix interface appeared uninfluenced by the moisture absorption and high temperatures. Zhou et al. [37] studied the bond durability of GFRP bars in concrete under different environments, including the tap water, alkaline solution (pH = 13.5), acid solution (pH = 2), and ocean water for different exposure durations (30, 60, and 90 days) at 20 °C. It was reported that there was no bond degradation under the simulated environments except for the acid solution. Even more, Alves et al. [38] conducted experiments on GFRP-concrete elements under sustained and fatigue loading conditions, and stated that the FT cycles enhanced the bond strength between the sand-coated GFRP bar and concrete by approximately 40%.

2.2. Statement of the problem

Clearly, although extensive studies have been carried out on the bond durability of the GFRP bars to concrete, the literature review generally demonstrates large discrepancies. This can be attributed to the different test methods and diversities in the characteristics of those test bars. Also, some laboratory tests considering the extreme environmental conditions may not correspond to field conditions that the structures actually experienced in reality. On the other hand, limited resources dedicated to GFRP reinforced structures exposed to aggressive cold environments including the combined effect of FT cycles, alkaline solution and saline solution. As such, the concrete pore solution that displays highly alkaline would be contaminated with chloride ions from de-icing salts, resulting in bond degradation of structures. Engineers need to comprehensively consider these environmental attacks on the long-term structural performance. Based on this, the objective of this study is to present an experimental study on the durability performance of the GFRP-concrete element, providing a reference and supplement to the current and future database for engineers and researchers.

3. Experimental program

3.1. Sample design

3.1.1. Material selection and sample size determination

The sample design was on the basis of a database generated in early work [1]. The database consisted of over 680 pullout specimens of GFRP bars in the literature. The critical information with reference to both GFRP bars and concrete mix can provide a sound basis for material selection and sample size determination.

Fig. 1 (a) and (b) show the surface treatment and diameter of GFRP bars used in previous studies. Failure modes and surface conditions are assigned with the legends. For simplicity, the first term of the legends used represents the failure modes, while the second term is for surface conditions: a) R = ribbed; b) HW = helically wrapped; c) SC = sand coated; d) HWSC = helically wrapped and sand coated; e) SW = spirally wrapped. Clearly, the ribbed surface, and helically wrapped and sand coated surface, both associated with pullout failure take up a large proportion among all surface types. Also, the bar diameter (d_b) ranging from 12 to 14 mm were mostly found in those pullout tests. Therefore, the GFRP bars having a nominal diameter of 12.7 mm and helically wrapped and sand-coated surface can be selected without loss of generality.

On the other hand, Fig. 2 (a) − (c) demonstrate the critical statistics for pullout specimen design in terms of concrete compressive strength (f'_c), bar embedment length (l_d) and concrete cover (c). First, Fig. 2 (a) indicates that the concrete compressive strength of about 40–50 MPa associated with pullout failure was the most of previous cases. Accordingly, the concrete compressive strength was reasonably designed to fall into the interval of [40,50] MPa with appropriate mix proportions. Second, Fig. 2 (b) illustrates that the embedment length to bar diameter ratio ranging from 5 to 6 is the most used parameter in the previous studies. Thus, the embedment length of GFRP bars was determined to be $l_d = 5d_b$, which is also in accordance with the pullout testing of FRP bars bonded to concrete stipulated in ASTM D 7913/D 7913M [39]. Finally, Fig. 2 (c) reveals that $c/d_b \geq 4$ associated with pullout failure accounts for the majority over other scenarios. This also conforms to the limit of 3.5 specified in ACI 440.1R-06 bond equation [40] when pullout failure is predicted. Upon this, the ratios of c/d_b with 1.5, 3.0 and 4.5 can be designed to allow different failure mode occurring.

3.1.2. Material properties

3.1.2.1. GFRP bars. The brand Aslan 100 series GFRP bars were used in this study. The surface of the GFRP bars is helically wrapped with fiber strands to create indentations along the bar, and sand particles are coated on the surface to enhance the bond strength, as shown in Fig. 3. All the bars are made of continuous longitudinal E-glass fibers impregnated in a thermosetting vinyl ester resin, and are manufactured using the pultrusion process. The nominal diameter of the bars is 12.7 mm. The tensile strength reduction from ambient at −40 °C is less than 5% according to ASTM D 7205 [41], and tensile strength retention due to the exposure to 12.8 pH solution for 90 days at 60 °C is greater than 80% according to ASTM D 7205 [41]. The 24 h moisture absorption at 50 °C is less than or equal to 0.25% according to ASTM D 570 [42]. The detailed mechanical and physical properties as reported by the manufacturer are summarized in Table 1.

3.1.2.2. Concrete. Since the bar diameter of 12.7 mm was selected, the concrete prism in terms of the concrete cover should allow both splitting failure and pullout failure to take place. The sample size of 127 × 127 × 177.8 mm was implemented by a set of formwork constructed out of plywood, as shown in Fig. 7. Prior to casting, the GFRP bars with bond breakers made of plastic tubes were deployed with different c/d_b of 1.5, 3.0 and 4.5, respectively (see Fig. 4). The embedment length was determined to be $5d_b = 63.5$ mm, which was generally assumed to be capable of representing the local bond behavior of the GFRP bar-concrete interface.

The concrete was prepared in the laboratory, with detailed mix proportions as presented in Table 2. Type I Portland cement was used for concrete mixing. The nominal maximum size of the coarse aggregate was 12.7 mm, and the fine aggregate size ranged from 0 to 4.75 mm. The concrete was mixed in a concrete mixer with an capacity of 0.03 m^3. In order to ensure mixing quality, totally three batches of concrete were used, of which the volume of each batch was less than 0.02 m^3. The coarse and fine aggregates were first mixed. Then the cement was added and thoroughly mixed. Finally, the water was added and the mixture was continuously mixed until the concrete exhibited uniform in appearance. The concrete was cast with GFRP bars in the horizontal position. After molding, all the samples were immediately covered with a plastic sheet to prevent moisture loss for 24 h, and then removed from the molds and cured in water at room temperature of 25 °C for 28 days. The mean concrete compressive strength of the specimens used in the test was determined according to ASTM C 39/C 39 M [43], where three 150 × 300 mm cylinders were used for each batch, as presented in Table 3.

3.2. Environmental aging design

3.2.1. Freeze-thaw cycles

The rapid FT cabinet (model G-118-H-3185B) was used to implement the repeated FT cycles on the test specimens in water. The original system is designed to accommodate up to eighteen

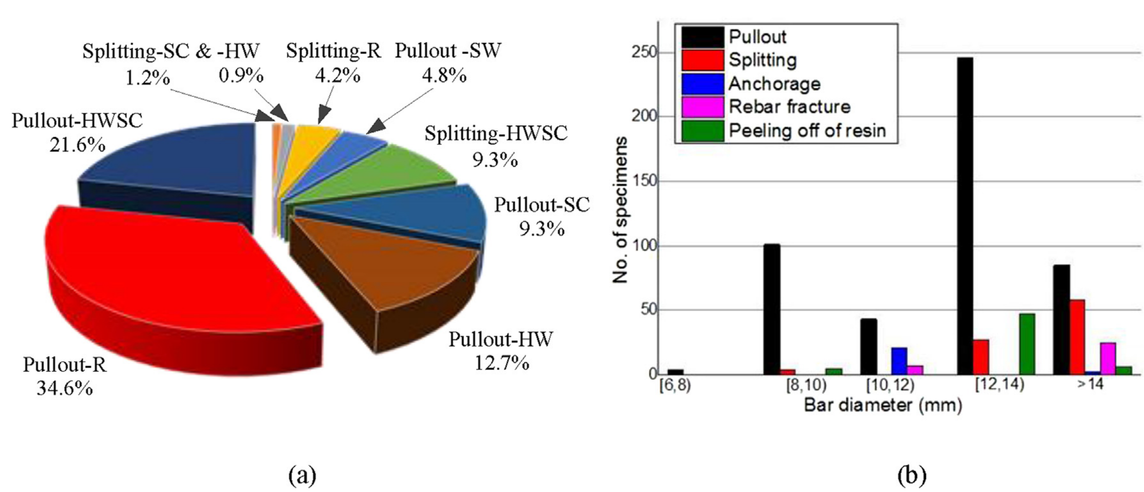


Fig. 1. General information of GFRP bars used in previous studies: (a) surface treatment and (b) bar diameter [1].

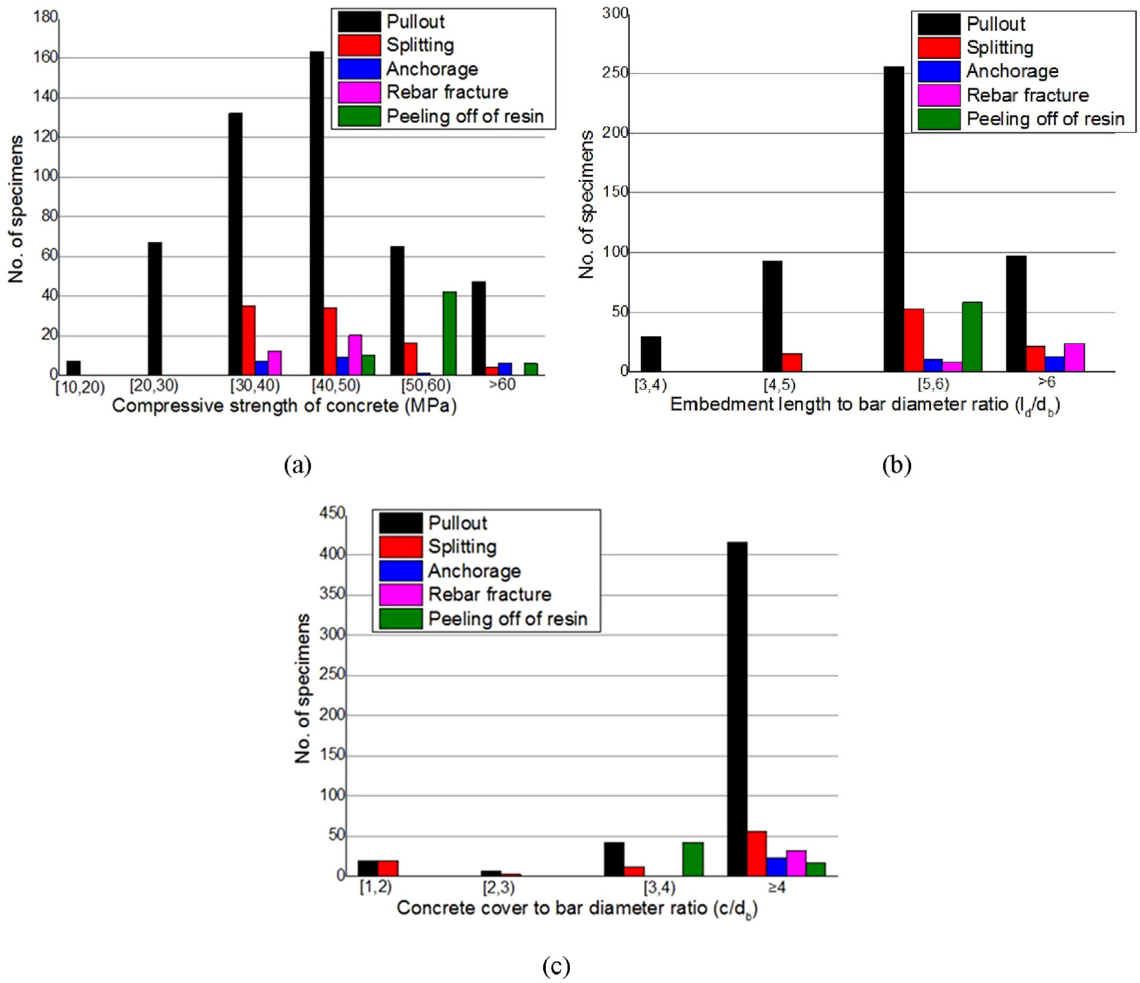


Fig. 2. General information of concrete mix and GFRP bars used in previous studies: (a) concrete compressive strength, (b) embedment length, and (c) concrete cover [1].

Fig. 3. GFRP bars used in this study.

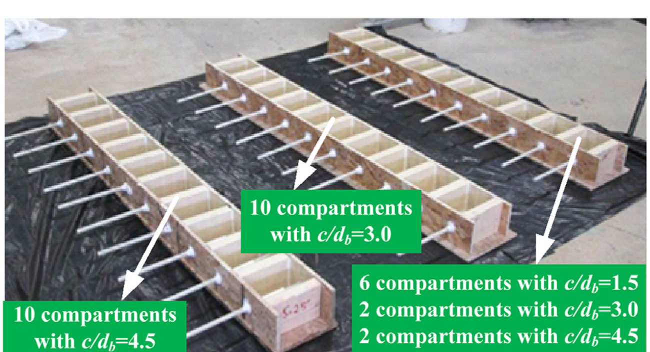


Fig. 4. View of the formwork manufactured with different c/d_b.

76 × 102 × 406 mm concrete prism specimens simultaneously, with one being a control. Consider the specimen size used in this study, the container of each compartment needs to be redesigned for its capacity. A container with internal space of 198 × 165 × 508 mm was determined to be capable of accommodating two 127 × 127 × 177.8 mm GFRP-concrete specimens with sufficient water coverage according to ASTM C 666/C 666M [44]. The deployment of the total six compartments in the cabinet is shown in Fig. 5.

The FT cycles were carried out according to ASTM C 666/C 666M Procedure A. The minimum and maximum core temperature of each specimen was set to be -18 ± 2 °C and 4 ± 2 °C. The control specimen was used to monitor and accurately record the complete temperature variations throughout the testing period. Each FT cycle was found to accomplish in approximately 5 h, and 40% of the time was used for thawing within each cycle. The specimens were immersed in water for at least 24 h before placing into the FT cabinet, and were taken out at the half and the end of the testing cycle in thawing condition for measurements. The weights of all the specimens before and after 75 FT cycles were measured to investigate the damage to concrete. The fundamental transverse and longitudinal frequencies of the concrete prism were measured at 0, 35 and 75 FT cycles for the purpose of calculating the dynamic modulus of elasticity according to ASTM C 215 [45], where the forced resonance method was used.

In this study, the test apparatus Humboldt H-3175 sonometer

Table 1
Material properties of 12.7 mm diameter GFRP bars (as reported by manufacturer).

	Item	Unit	Value
Mechanical properties	Guaranteed tensile strength	MPa	758
	Tensile modulus of elasticity	GPa	46
	Ultimate strain	%	1.64
	Strength retention due to alkali resistance	%	>80
	Strength reduction at cold temperature	%	<5
Physical properties	Transition temperature of Resin	°C	>110
	Moisture absorption	%	≤0.25
	Glass fiber content by weight	%	>70

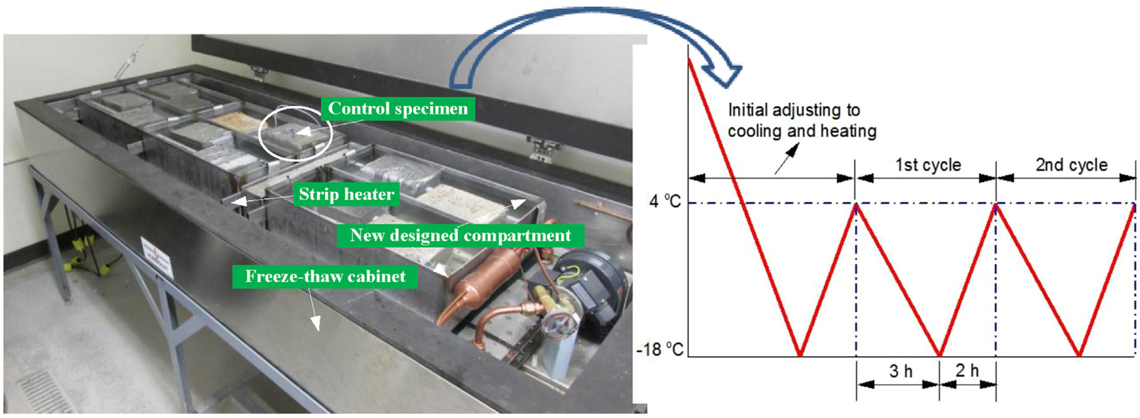


Fig. 5. Demonstration of the freeze-thaw cycling test.

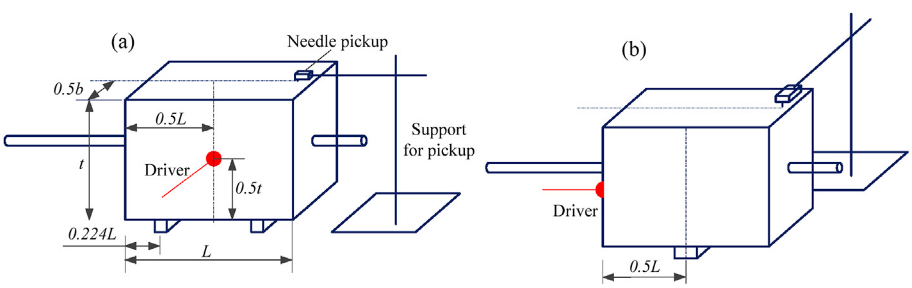


Fig. 6. Demonstration of dynamic modulus test for (a) transverse mode and (b) longitudinal mode.

utilizing a phono-type cartridge as a pickup was adopted to determine changes in resonance frequency of the concrete specimens. The driver and pickup are mounted on portable stands, which allows for greater flexibility in testing. The basic testing schematic is to vibrate the supported specimen using a driving unit with varying frequencies and record the response using a lightweight pickup unit. The value of frequency was recorded as the resonance frequency when the measured response reaches the maximum amplitude.

Fig. 6 (a) demonstrates the test setup for the specimen that is able to vibrate freely in transverse mode. The specimen is supported at 0.224*L* from the edges, and the driving unit is placed at the center of one surface, producing mechanical vibrations and imparting these vibrations to the test specimen. Meanwhile, a lightweight pickup unit is set at the middle along the width of the specimen. Fig. 6 (b) shows the test setup for the longitudinal mode, where the specimen is supported at the middle span, and the driving unit is placed approximately at the center of one end surface to produce longitudinal excitation to the concrete prism. For both testing procedures, the frequency of vibrations can be controlled ranging from 400 to 12,000 cycles per second with an accuracy of better than ±2%. The driving frequency is adjusted gradually until a maximum response is captured and displayed on a built-in digital counter, which is the resonant frequency of the specimen.

The dynamic modulus can be calculated based on the respective fundamental frequencies as shown in Eqns. (1) and (2),

$$E = CMn^2 \tag{1}$$

or

$$E = DM(n')^2 \tag{2}$$

where E is the dynamic modulus of elasticity; $C = 0.9464\,(TL^3/bt^3)$, T is the correction factor and calculated to be 3.58 according to ASTM C 215 [45], M is the mass of the specimen in kg; $D = 4\,(L/bt)$, n' is the fundamental longitudinal frequency in Hz.

3.2.2. Alkaline-saline solution

The alkaline-saline (AS) solution was considered to simulate the concrete pore solution contaminated with chloride ions from de-

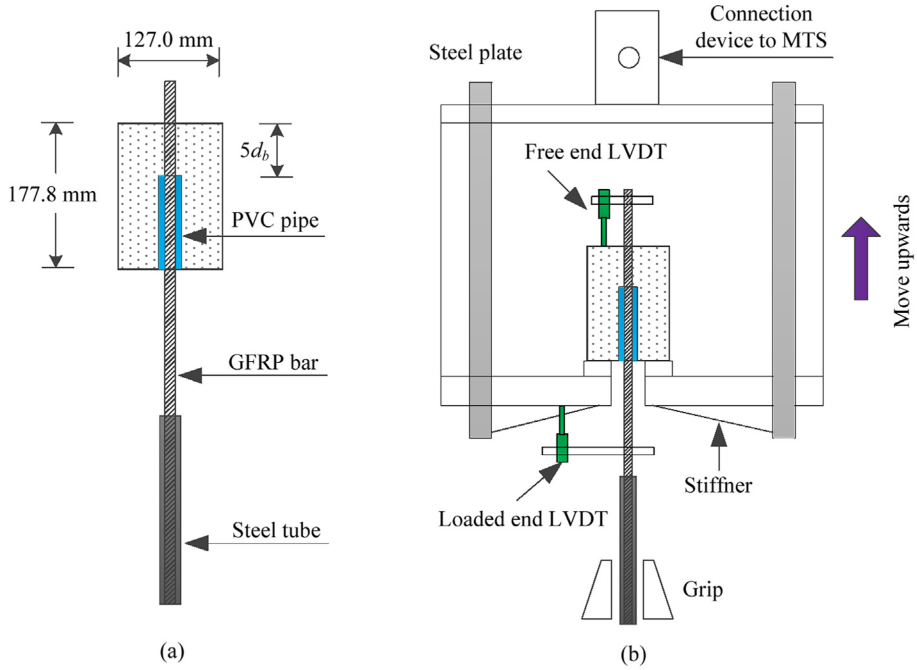


Fig. 7. Schematic demonstration of the (a) pullout specimen and (b) test setup.

Table 2
Composition and characteristics of concrete.

Item	Unit	Value
Water (W)	kg/m^3	186
Cement (C)	kg/m^3	503
W/C	–	0.38
Coarse aggregate	kg/m^3	910
Fine aggregate	kg/m^3	717
Air-entraining agent	mL/m^3	312
28-day compressive strength	MPa	34 ± 1
Air content	%	5.2

icing salts. The alkaline solution with a pH = 12.5 was prepared with a mix of sodium hydroxide (NaOH), calcium hydroxide ($Ca(OH)_2$) and potassium hydroxide (KOH) in the ratio of 2.4:2:19.6 g/L, respectively [21]. The 3% concentration sodium chloride (NaCI) was added into the alkaline solution to implement the AS solution [7,30]. In addition to the room temperature (25 °C), the elevated temperature of 90 °C was used to accelerate the degradation rate of GFRP-concrete elements for duration of 90 days before testing.

3.3. Pullout test design

The pullout tests were performed using the universal testing machine that is capable of handling loads up to 1000 kN. The GFRP-concrete specimens were sheathed with thick-wall hollow steel pipes at the free portion of the GFRP bar, as shown in Fig. 7 (a). The pipes were 203 mm in length and 19.05 mm in inside diameter, which were further processed with internal threaded along the length to increase the roughness of anchorage. The pipes were filled with a commercial epoxy to adhere the bar for at least 24 h to take effect before testing. The schematic demonstration of the test setup is shown in Fig. 7 (b). The sample encased in a steel reaction frame was instrumented with three linear variable differential transformers (LVDTs) to record the elongation of the bar. All the tests were conducted using displacement control at a rate of 0.02 mm/s, thus the post peak behavior of the bond-slip relationship can be obtained. The load was measured with the electronic loading cell of the machine, and the slips at both loaded end and free end were measure with the LVDTs. All measurements, including the pullout load and displacements, were recorded synchronously by an automatic data-acquisition system at a rate of 2 data/s.

4. Experimental results and discussions

A total of 26 GFRP-concrete specimens were considered in this test, including 3 specimens with $c/d_b = 1.5$, 11 specimens with $c/d_b = 3.0$, and 12 specimens with $c/d_b = 4.5$ to investigate the bond durability under different environmental conditions. The specimen identification is in the form: M-G-C-N, where M denotes the batch of the concrete mix, G denotes the specimen group, C denotes the concrete cover to bar diameter ratio, and N denotes the specimen number. For example, M3-D-4.5–2 indicates that the specimen No.2 with $c/d_b = 4.5$ was derived from concrete mix batch No.3, and was conditioned with the coupled FT cycles and AS solution. In the pullout test, the stress along the embedment length is not constantly distributed and hence, the average bond stress is defined as:

$$\tau = \frac{P}{\pi d_b l_d} \tag{3}$$

where P is the tensile load. The experimental results obtained from the pullout test were detailed in Table 3, where $P_{\max}$ is the maximum tensile load, $\tau_{\max}$ is the bond strength, $\tau^r_{\max}$ is the bond strength retention, $s_{m,le}$ and $s_{m,fe}$ are the slips corresponding to bond strength at the loaded end and free end, respectively. The mean values of the bond strength and corresponding slips of nominally identical specimens are also listed. Meanwhile, the normalized bond strength $\tau^*_{\max}$ is used to account for the influence of concrete compressive strength, which is defined as:

Table 3
Pullout test results of GFRP-concrete specimens exposed to different environmental conditions.

Specimen	f'_c (MPa)	P_{max} (kN)	τ_{max} (MPa)	τ_{max}[a] (MPa)	$s_{m,le}$ (mm)	$s_{m,le}$[a] (mm)	$s_{m,fe}$ (mm)	$s_{m,fe}$[a] (mm)	τ^*_{max} (MPa$^{0.5}$)	τ^r_{max}[a] (%)	Failure mode[b]
Unconditioned (control) specimens: Group A											
M1-A-1.5-1	44.22	29.11	11.49	13.03	0.39	0.50	N/A	0.15	1.73	100	S
M1-A-1.5-2	44.22	37.42	14.77		0.62		0.13		2.22		S
M1-A-1.5-3	44.22	32.48	12.82		0.48		0.17		1.93		S
M2-A-3.0-1	43.09	43.73	17.26	17.30	1.78	1.71	1.22	1.26	2.63	100	PO
M2-A-3.0-2	43.09	45.71	18.04		1.69		1.40		2.75		PO
M2-A-3.0-3	43.09	42.03	16.59		1.67		1.15		2.53		PO
M3-A-4.5-1	46.73	48.42	19.11	18.86	1.53	1.42	0.67	0.65	2.80	100	PO
M3-A-4.5-2	46.73	47.05	18.57		1.32		0.65		2.72		PO
M3-A-4.5-3	46.73	47.86	18.89		1.41		0.64		2.76		PO
Specimens conditioned with freeze-thaw cycles: Group B											
M2-B-3.0-1	43.09	40.38	15.94	15.31	2.13	1.99	1.44	1.41	2.43	88	S
M2-B-3.0-2	43.09	37.19	14.68		1.85		1.38		2.24		S
M3-B-4.5-1	46.73	45.78	18.07	18.27	1.75	1.72	1.15	1.17	2.64	97	PO
M3-B-4.5-2	46.73	46.64	18.41		1.79		1.22		2.69		PO
M3-B-4.5-3	46.73	46.44	18.33		1.61		1.14		2.68		PO
Specimens conditioned with alkaline-saline solution: Group C											
M2-C-3.0-1	43.09	44.34	17.50	17.17	1.95	1.92	1.54	1.53	2.67	99	PO
M2-C-3.0-2	43.09	43.60	17.21		1.89		1.48		2.62		PO
M2-C-3.0-3	43.09	42.54	16.79		1.91		1.57		2.56		PO
M3-C-4.5-1	46.73	44.89	18.72	18.58	1.67	1.67	0.62	0.62	2.59	94	PO
M3-C-4.5-2	46.73	45.81	18.68		1.61		0.59		2.64		PO
M3-C-4.5-3	46.73	43.91	18.33		1.72		0.66		2.54		PO
Specimens conditioned with both freeze-thaw cycles and alkaline-saline solution: Group D											
M2-D-3.0-1	43.09	37.62	14.85	14.87	2.03	1.95	1.74	1.69	2.26	86	S
M2-D-3.0-2	43.09	38.26	15.10		1.97		1.69		2.30		S
M2-D-3.0-3	43.09	37.14	14.66		1.86		1.64		2.23		S
M3-D-4.5-1	46.73	43.37	17.12	17.34	1.81	1.78	1.35	1.38	2.50	93	PO
M3-D-4.5-2	46.73	44.49	17.56		1.89		1.38		2.57		PO
M3-D-4.5-3	46.73	43.96	17.35		1.65		1.40		2.54		PO

[a] Mean value for nominally identical specimens.
[b] PO = pullout failure; S = splitting failure.

$$\tau^*_{max} = \frac{\tau_{max}}{\sqrt{f'_c}} \tag{4}$$

Table 3 details the mode of failure of unconditioned and conditioned pullout specimens. Splitting failure took place in all the control specimens with concrete cover $c = 1.5d_b$, whereas pullout failure occurred in the other control specimens with $c = 3.0d_b$ and $c = 4.5d_b$. Thus, the conditioned specimens that exposed to individual FT cycles and AS solutions, as well as the coupled effect of the both, considered these two types of concrete covers to investigate the environmental attack on the failure mode.

Fig. 8 (a) shows the typical splitting failure and its corresponding interface between the GFRP bar and concrete for both unconditioned and conditioned specimens. This failure mode demonstrates obvious cracks that initiated from the GFRP-concrete interface and further developed up to the concrete outer surface, leading to brittle failure during the pullout process. The conditioned specimens with $c = 3.0d_b$ experienced to individual FT cycles and coupled FT cycles and AS solution all failed by concrete splitting rather than by pullout of the bar. Such change of failure mode indicates the concrete cover $c = 3.0d_b$ is not capable of resisting attacks of environmental agents including FT cycles, whereas transverse reinforcements are required to provide additional constraint to concrete. This conclusion also confirms the limit of $c = 3.5d_b$ term stipulated in ACI design codes [40] when the pullout failure is usually predicted. The GFRP bars in these specimens exhibited minor scratches on the bar surface along the embedment length (e.g., specimen M2-D-3.0-1). The sands and fiber strands remained intact and closely adhere to the bar surface. On the contrary, the specimens conditioned with individual AS solutions all failed by pullout of the bars for both $c = 3.0d_b$ and $c = 4.5d_b$. The individual AS solution did not change the failure mode. Thus, the FT cycles play more significant influence on the bond durability of GFRP-concrete element, which may result in brittle failure when the concrete cover is not sufficient to hold reinforcements.

On the other hand, the specimens failed by pullout failure displayed similar conditions at the bar-concrete interface, as shown in Fig. 8 (b). The sand particles and wrapped strands significantly peeled off due to the friction and sliding of pulling, with residues being observed at the concrete interface along the embedment length (e.g., specimen M3-A-4.5–3). The conditioned specimens showed similar phenomenon as that presented in the unconditioned specimen. There were no trace of chemical attack in terms of abnormal color and corrosive substances being observed except for the stripping of sand residues on the concrete interface or removal of the resin. The specimens with concrete cover $c = 4.5d_b$ all failed by pullout failure in those conditioned with individual FT cycles, AS solution, and coupled effect of both, indicating that $c = 4.5d_b$ is capable of preventing brittle failure under those environmental attacks.

4.1. Weight loss

Scaling is a degradation phenomenon of hardened concrete, of which a local peeling or flaking of a finished surface takes place due to environmental attacks. The deterioration of concrete structures is usually related to surface scaling, especially when they are subjected to FT cycles. It usually initiates from localized small patches that may further extend to expose large areas. The mortar and paste strip from the concrete surface or the alkali-silica reacts with the alkali-reactive aggregate inside the concrete mix, leading to loosening of the coarse aggregates and ultimate strength reduction of concrete structures.

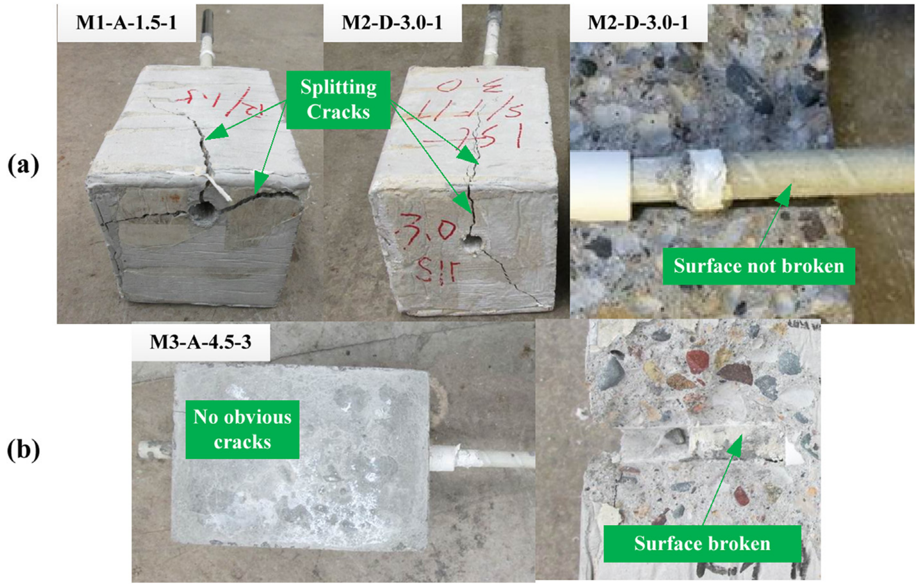


Fig. 8. Typical failure modes for unconditioned and conditioned specimens: (a) splitting and (b) pullout failures.

Fig. 9 shows the surface conditions of the specimens exposed to individual environment of FT cycles, AS solution, and coupled effect of the both. The control specimen M3-A-4.5-1 displayed intact surface in contrast to the conditioned specimens. The specimen M3-C-4.5−1 that was conditioned with AS solution exhibited yellowish white color and rough mortar surface whereas no obvious peeling of patches and cracks were observed. The specimens conditioned with respective FT cycles and coupled FT cycles and AS solution demonstrated different degrees of surface scaling. The specimen M3-B-4.5-3 showed peeling of the surface mortar at the edge of the prism especially near the arris of the outer surface whereas no obvious aggregates exposed. The specimen M3-D-4.5-2 clearly displayed the coarse aggregate with distinct loss of the surface mortar, indicating the couple environments of FT cycles and AS solution may lead to more severe deterioration to concrete integrity.

The weight loss of all the conditioned specimens was measured using Eqn. (5),

$$\Delta W_n = \frac{W_0 - W_r}{W_r} \times 100 \tag{5}$$

where ΔW_n is the weight loss after n FT cycles, W_0 and W_r are the original weight and residual weight of the specimen before and after conditioning, respectively. More detailed information is summarized in Table 4 and plotted in Fig. 10, where characters located before and after the hyphen denote the environmental conditions and concrete cover to bar diameter ratio, respectively. For example, FT + AS-3.0 indicates the specimen with $c = 3.0d_b$ was experienced coupled environments of FT cycles and AS solution. The specimens exposed to AS solutions showed the smallest weight loss among other environments. The mean weight losses were 0.24% and 0.21% for the specimens with $c = 3.0d_b$ and $c = 4.5d_b$, respectively. The specimens subjected to FT cycles lost less than 1% of their original weight, with 0.70% and 0.72% for the specimens with $c = 3.0d_b$ and $c = 4.5d_b$, respectively. This indicates that the air-entrained concrete performed well according to ASTM C 666/C 666M Procedure A [44]. However, when the specimens were experienced with coupled environments of FT cycles and AS solutions, the weight loss increased to 1.23% and 1.34% for specimens with $c = 3.0d_b$ and $c = 4.5d_b$, respectively. The coupled environments resulted in the largest weight loss of the GFRP-concrete specimens among all the environmental conditions.

Considering that field concrete structures in cold regions inevitably experience the concrete pore solution contaminated with de-

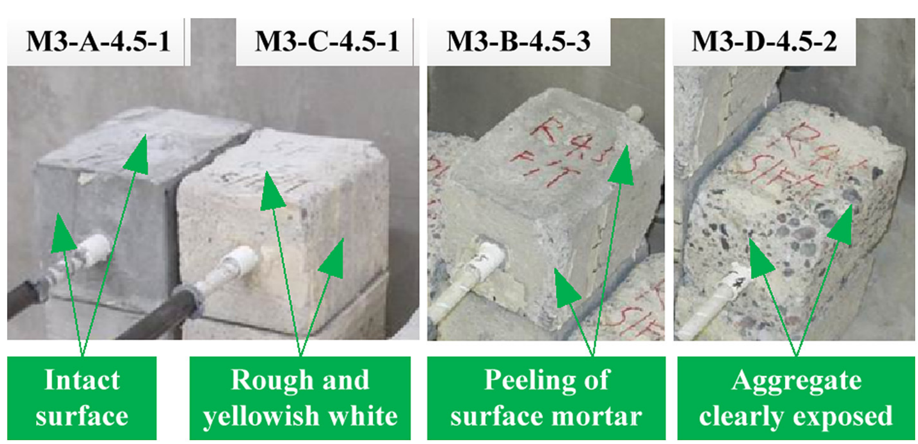


Fig. 9. Examples of specimens exposed to different environmental conditions.

Table 4
Weight loss of conditioned specimens.

Specimen	Original weight (g)	Residual weight (g)	Weight loss (%)	Mean (%)	Standard deviation (%)
Freeze thaw cycles					
M2-B-3.0-1	6706.2	6669.3	0.55	0.70	0.14
M2-B-3.0-2	6710.0	6654.3	0.83		
M3-B-4.5-1	6415.2	6363.9	0.80	0.72	0.09
M3-B-4.5-2	6566.3	6518.4	0.73		
M3-B-4.5-3	6670.5	6629.1	0.62		
Alkaline-saline solution					
M2-C-3.0-1	6608.6	6592.1	0.25	0.24	0.09
M2-C-3.0-2	6559.7	6550.5	0.14		
M2-C-3.0-3	6616.2	6594.4	0.33		
M3-C-4.5-1	6600.3	6590.4	0.15	0.21	0.05
M3-C-4.5-2	6679.8	6663.1	0.25		
M3-C-4.5-3	6547.0	6531.9	0.23		
Coupled freeze-thaw cycles & alkaline-saline solution					
M2-D-3.0-1	6686.7	6604.5	1.23	1.23	0.20
M2-D-3.0-2	6702.1	6606.3	1.43		
M2-D-3.0-3	6627.6	6559.3	1.03		
M3-D-4.5-1	6558.5	6477.2	1.24	1.34	0.12
M3-D-4.5-2	6563.8	6477.8	1.31		
M3-D-4.5-3	6560.4	6464.6	1.46		

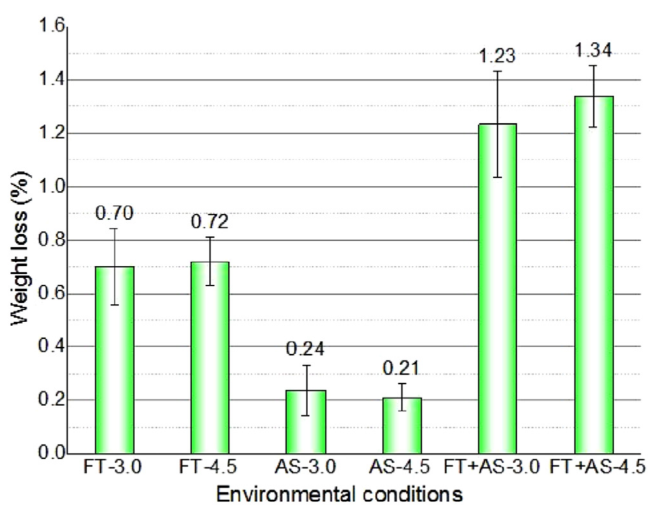


Fig. 10. Weight loss under different environmental conditions.

icing salts and freeze-thaw cycles due to climate change. Thus, the conclusions drawn from the simulated environmental conditions reveal that the coupled FT cycles and AS solution may be more hazardous than the individual FT cycles despite the results obtained in the laboratory test may be more severe due to higher cycling rate or deviations with the practical situation.

4.2. Relative dynamic modulus of elasticity and durability factor

The specimens were measured after 35 and 75 FT cycles respectively to investigate the potentially deteriorating influences. Considering that the dynamic modulus of different specimens may vary from each other due to different mix batch or aggregate distribution. Thus, the relative dynamic modulus of elasticity is defined in the following according to ASTM C 666/C 666M [44]:

$$P_n = \frac{E_n}{E_0} \times 100 \tag{6}$$

where P_n is the relative dynamic modulus of elasticity, E_0 and E_n are the dynamic modulus of elasticity at 0 and n FT cycles. It aims to facilitate the comparisons among different individuals, eliminating the influences caused by the diversity of the test specimens. Table 5 details the dynamic modulus of elasticity calculated based on respective transverse and longitudinal vibration modes. Generally, the dynamic modulus calculated using the transverse vibration mode was slightly smaller than that using the longitudinal vibration mode, which is consistent with the observation in Ref. [46]. The maximum error of the dynamic modulus using different vibration modes was 3.94%, indicating the forced resonance method performed in this study yielded good accuracy.

Fig. 11 shows the comparisons of the relative dynamic modulus of elasticity after 35 and 75 FT cycles. Taking the examples based on the transverse vibration mode, in general, the specimens experienced coupled FT cycles and AS solution displayed smaller relative dynamic modulus of elasticity than those experienced individual FT cycles. The R_E of the specimens M2-B-3.0, M3-B-4.5 decreased to the respective percent of 93.70 and 83.28 after 35 FT cycles, and further decreased to the respective percent of 73.80 and 69.68 after 75 FT cycles. While the R_E of the specimens M2-D-3.0 and M3-D-4.5 decreased to the respective percent of 84.17 and 78.54 after 35 FT cycles combined with AS solution, and further decreased to the respective percent of 58.32 and 61.48 after 75 FT cycles combined with AS solution. In particular, the largest reduction in R_E, 25.85% between 35 FT cycles combined with AS solution and 75 FT cycles combined with AS solution, was observed in the specimen M2-D-3.0 among all scenarios, indicating that smaller concrete cover may suffer from more potentially deteriorating influences as the number of FT cycles increases. Such phenomenon is consistent with the splitting failure observed in the specimens M2-B-3.0-1, M2-B-3.0-2, M2-D-3.0-1, M2-D-3.0-2, and M2-D-3.0-3, of which the smaller concrete cover resulted in weaker FT resistance.

In addition, the durability factor is defined according to ASTM C 666/C 666M [44],

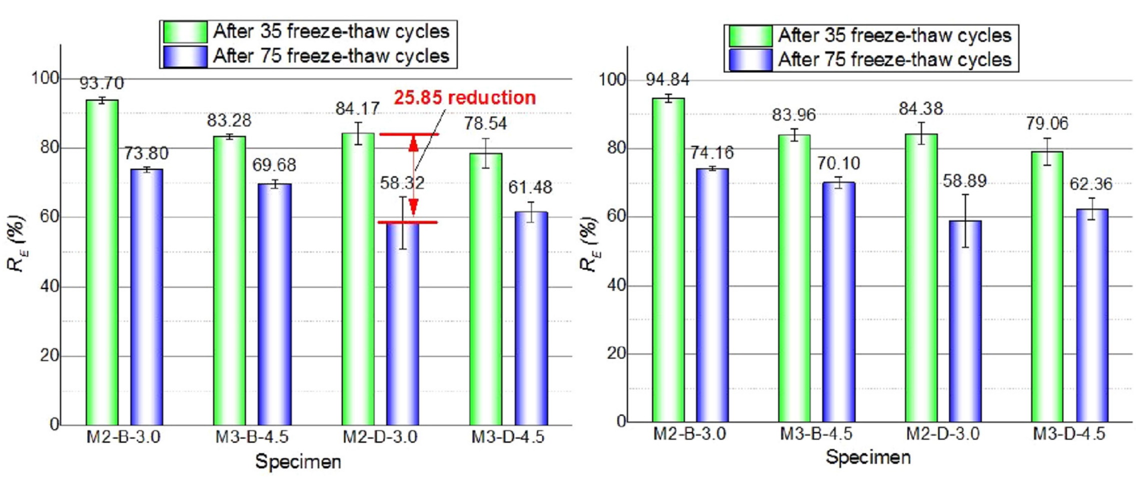
$$DF = P_n \cdot n/m \tag{7}$$

where DF is the durability factor of the test specimen; P_n is the relative dynamic modulus of elasticity at n cycles in %, and the transverse vibration mode is used herein; and m is the specified number of cycles at which the exposure is to be terminated, which is 75 since the FT cycles was conducted to 75 cycles in this study.

Table 5
Relative dynamic modulus of elasticity.

Specimen	Transverse vibration mode				Longitudinal vibration mode				
	E (GPa)	R_E (%)	Mean (%)	Standard deviation (%)	E (GPa)	R_E (%)	Mean (%)	Standard deviation (%)	Error[a] (%)
Before conditioned									
M2-B-3.0-1	71.32	100	100	0	72.11	100	100	0	1.11
M2-B-3.0-2	75.36	100			76.13	100			1.02
M3-B-4.5-1	65.89	100	100	0	66.22	100	100	0	0.50
M3-B-4.5-2	64.81	100			65.58	100			1.19
M3-B-4.5-3	69.19	100			70.67	100			2.14
M2-D-3.0-1	72.44	100	100	0	73.08	100	100	0	0.88
M2-D-3.0-2	74.69	100			75.45	100			1.02
M2-D-3.0-3	76.18	100			77.02	100			1.10
M3-D-4.5-1	69.02	100	100	0	69.88	100	100	0	1.25
M3-D-4.5-2	64.51	100			65.46	100			1.47
M3-D-4.5-3	67.67	100			68.39	100			1.06
After 35 freeze thaw cycles									
M2-B-3.0-1	67.26	94.31	93.70	0.86	69.01	95.70	94.84	1.22	2.60
M2-B-3.0-2	70.15	93.09			71.54	93.97			1.98
M3-B-4.5-1	55.36	84.02	83.28	0.75	57.03	86.12	83.96	1.87	3.02
M3-B-4.5-2	53.48	82.52			54.28	82.77			1.50
M3-B-4.5-3	57.63	83.29			58.65	82.99			1.77
After 35 freeze-thaw cycles & alkaline-saline solution									
M2-D-3.0-1	63.58	87.77	84.17	3.13	64.37	88.08	74.16	0.56	1.24
M2-D-3.0-2	61.72	82.63			62.49	82.82			1.25
M2-D-3.0-3	62.55	82.11			63.35	82.25			1.28
M3-D-4.5-1	51.64	74.82	78.54	4.31	52.58	75.24	70.10	1.63	1.82
M3-D-4.5-2	53.71	83.26			54.61	83.42			1.68
M3-D-4.5-3	52.48	77.55			53.70	78.52			2.32
After 75 freeze thaw cycles									
M2-B-3.0-1	52.98	74.28	73.80	0.69	53.76	74.55	84.38	3.21	1.47
M2-B-3.0-2	55.25	73.31			56.15	73.76			1.63
M3-B-4.5-1	46.78	71.00	69.68	1.26	47.66	71.97	79.06	4.12	1.88
M3-B-4.5-2	44.39	68.49			45.22	68.95			1.87
M3-B-4.5-3	48.12	69.55			49.03	69.38			1.89
After 75 freeze-thaw cycles & alkaline-saline solution									
M2-D-3.0-1	47.83	66.03	58.32	7.47	48.72	66.67	58.89	7.56	1.86
M2-D-3.0-2	43.17	57.80			44.08	58.42			2.11
M2-D-3.0-3	38.94	51.12			39.73	51.58			2.03
M3-D-4.5-1	40.16	58.19	61.48	3.00	41.02	58.70	62.36	3.17	2.14
M3-D-4.5-2	41.33	64.07			42.16	64.41			2.01
M3-D-4.5-3	42.08	62.18			43.74	63.96			3.94

[a] Error is defined as the absolute difference between the E calculated using transverse and longitudinal modes divided by the E using the transverse mode.



(a) Transverse vibration mode (b) Longitudinal vibration mode

Fig. 11. Variations of relative dynamic modulus after 35 and 75 freeze-thaw cycles.

Fig. 12 demonstrates the impact of environmental conditions on the durability factor, where the specimens experienced individual 75 FT cycles and coupled 75 FT cycles and AS solution were used. It is clear that larger reductions in durability factor were observed when the specimens were subjected to the coupled environment rather than the individual environment. The durability factor of all

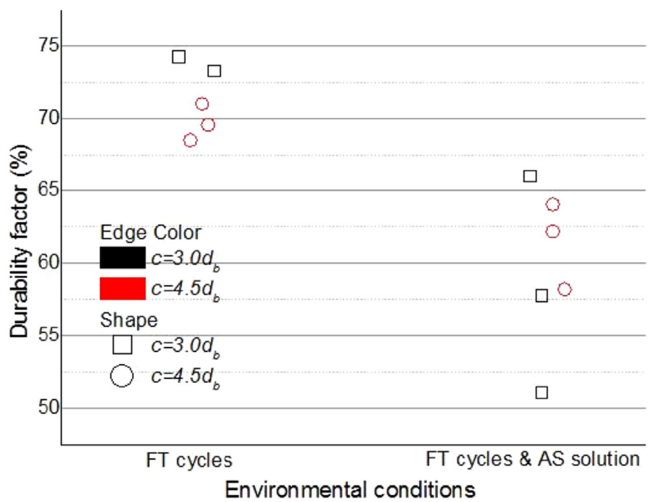


Fig. 12. Effect of environmental conditions on durability factor.

specimens experienced FT cycles were larger than 40% while smaller than 85%, indicating the medium FT resistance of the test specimens [47].

4.3. Bond behavior and durability

4.3.1. Bond stress-slip response

The bond stress-slip relationship at the loaded end and free end are shown in Fig. 13, where the representative of the control specimens and environmental conditioned specimens with different concrete covers are presented. Generally, the ascending branches of all the bond-slip curves at the free end displayed apparent lag effect compared to those at the loaded end, which conform to the fact that the slip development at the free end lags behind that at the loaded end. The descending branches associated with the conditioned specimens failed by concrete splitting (e.g., specimen M2-D-3.0-1) showed shorter segment compared with the specimens failed by the pullout of the bars (e.g., specimen M3-D-4.5−3). For the typical specimens M2-A-3.0-2, M2-B-3.0-1, M2-C-3.0-1, and M2-D-3.0-1, the complete bond-slip curves at both loaded end and free end of the conditioned specimens located below the bond-slip curves of the control specimens. This indicates that the specimens having smaller concrete cover $c = 3.0d_b$, may have lower bond resistance to environmental attacks at both pre-peak stage and post-peak stage of the bond development. On the other hand, for the typical specimens M3-A-4.5−2, M3-B-4.5−1, M3-C-4.5−1, M3-D-4.5−3, the ascending branches exhibited minor difference between the control specimen and conditioned specimens, whereas the descending branches presented obvious reduction in bond stress. This suggests that increasing the concrete cover may have great contribution in enhancing the pre-peak bond resistance to the environmental attacks in terms of AS solution, FT cycles and coupled influences of the both.

Compared to the control specimens, it can be noticed that the conditioned specimens associated with the coupled FT cycles and AS solution exhibited larger reductions in the bond stresses than those associated with the individual environmental agent. In particular, the specimens conditioned with the AS solution presented the smallest reduction among all the environmental scenarios, as demonstrated in the green dash lines of all the figures. Moreover, the impact of the individual FT cycles on the specimens having larger concrete cover mainly reflected in the post-peak bond behavior, as illustrated in the bond-slip curves of the specimen M3-B-4.5−1. Referring to the bond toughness introduced in Ref. [48], this indicates that the capacity of the post-peak energy absorption may decrease more obviously than the capacity of the pre-peak energy absorption when subjected to FT cycles.

4.3.2. Bond strength and its corresponding slip

The bond strengths of the specimens under different environmental conditions are displayed in Fig. 14 (a). In order to eliminate the influence of different concrete mix on the test results, the normalized values with respect to the square root of concrete compressive strength were presented in Fig. 14 (b). Generally, it is noticed that all the conditioned specimens showed deterioration in their bond strength. For the specimens having smaller concrete cover $c = 3.0d_b$, the environmental conditions including FT cycles exhibited more detrimental impact on the bond strength whereas the AS solution demonstrated minor influence. In particular, the coupled conditioning and individual FT cycles resulted in 14% and 12% reductions in both bond strength and normalized bond strength, respectively. While for the specimens having larger concrete cover $c = 4.5d_b$, the FT resistance was significantly improved accordingly. The bond strength reduction was reduced to 8% under the coupled conditioning, and 3% under the individual FT cycles. The normalized bond strength of all the conditioned specimens exhibited the similar patterns to the bond strength observation. It is worth noting that the coupled environment was the worst case regardless of the concrete cover.

The slips corresponding to the bond strength at the loaded end and free end are presented in Fig. 15. The results reveal that the specimens having larger concrete cover exhibited smaller slip at both loaded end and free end compared to those having smaller concrete cover. The conditioned specimens generally showed larger slip than the unconditioned specimens despite the discrepancy occurred due to the random effect of individual test results, which was consistent with the observations in Ref. [49]. In particular, the loaded end slip has more significance in practice. The slip associated with the FT cycles decreased from 1.99 to 1.72 mm when the concrete cover increased from $c = 3.0d_b$ to $c = 4.5d_b$, with approximately 14% reduction due to the enhanced confinement to the GFRP bars. Such test results were consistent with the failure mode observed in these two different concrete covers. Moreover, the slip of the specimens ($c = 4.5d_b$) conditioned with the coupled environment was 1.78 mm with approximately 20% increase compared with the control specimen.

5. Calibration of analytical models considering environmental effects

Currently, the FRP-concrete bond under pullout loads can be demonstrated through several available analytical models, which use a set of explicit expressions to depict the development of the bond stress against slip [50–52]. The model proposed by Malvar [50] uses a polynomial function in terms of seven curve-fitting parameters to describe the bond-slip relationship. Nevertheless, the ascending branch was later evaluated to be less reliable for FRP material [53]. The BPE model developed by Eligehausen et al. [54] was originally used to illustrate the steel-concrete bond, and then applied to FRP bars by recalibrating those parameters [51,52,55]. Cosenza et al. [51] and Rossetti et al. [52] conducted experiments of FRP bars with different surface treatments. However, their test results were too scattered to determine the parameters. More detailed comparisons among those analytical models can be found in Refs. [56] and [57]. This study adopted another two widely used analytical models viz., modified Bertero-Eligehausen-Popov (mBPE) model and Cosenza-Manfredi-Realfonzo (CMR) model for the bond-slip prediction. Moreover, considering that the calibrated

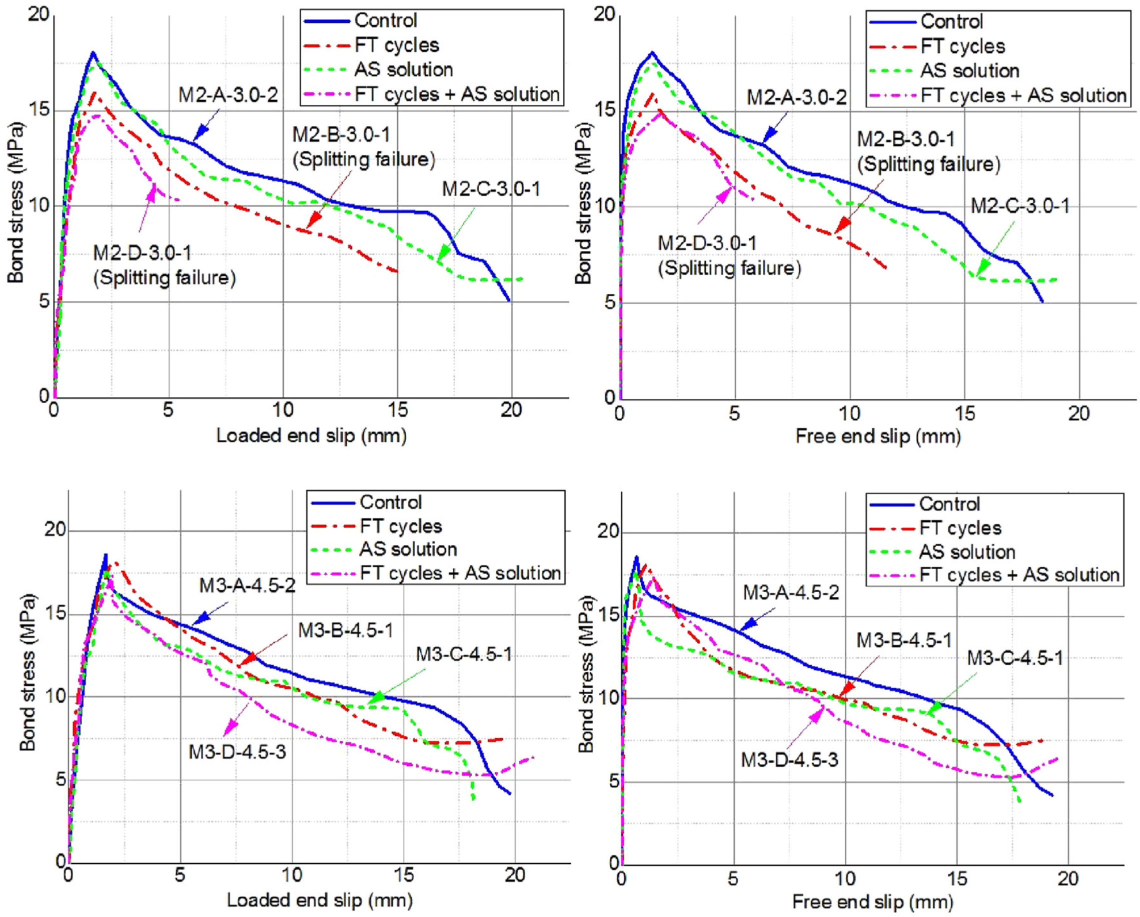
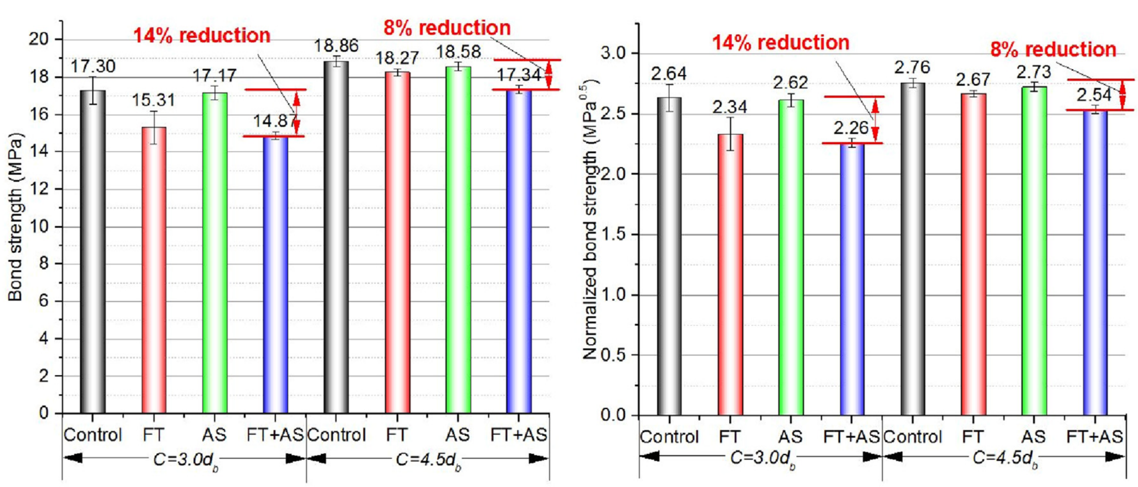


Fig. 13. Typical bond stress-slip relationship for unconditioned and conditioned specimens.



(a) (b)

Fig. 14. (a) Bond strength and (b) normalized bond strength.

models in previous studies did not consider the environmental impact on the bond performance and hence, it is necessary to provide a more accurate calibration involving the environmental influence to better demonstrate the GFRP- concrete bond behavior.

5.1. mBPE model

The bond-slip relationship of the mBPE model can be expressed as a piecewise function in Eqn. (8),

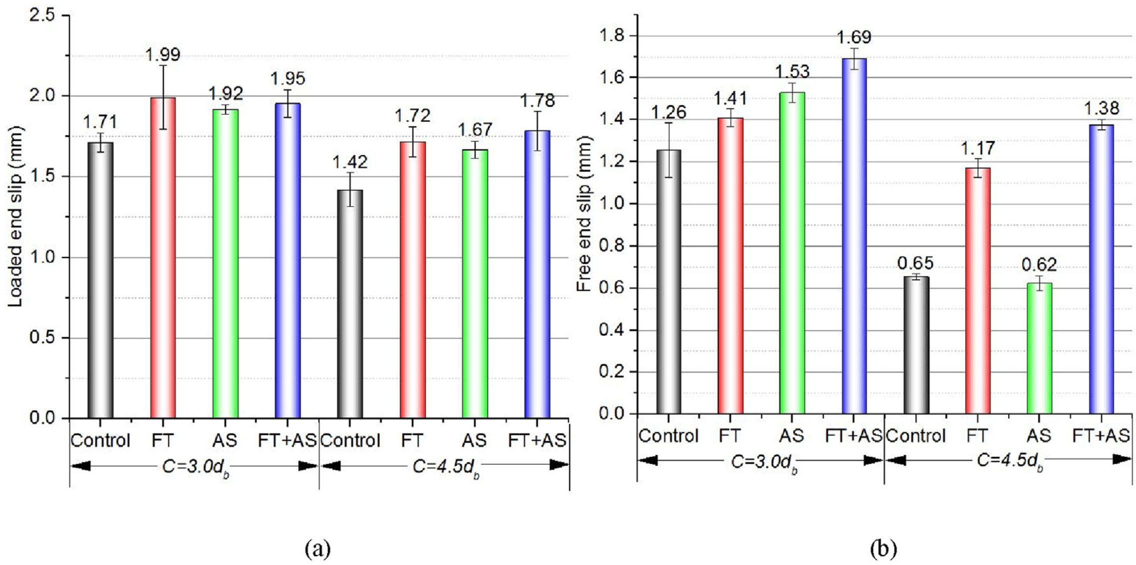


Fig. 15. Slip corresponding to bond strength at (a) loaded end and (b) free end.

$$\frac{\tau}{\tau_b} = \begin{cases} \left(\dfrac{s}{s_b}\right) & (0 \leq s \leq s_b) \\ 1 - p\left(\dfrac{s}{s_b} - 1\right) & (s_b \leq s \leq s_3) \\ \dfrac{\tau_3}{\tau_b} & (s \geq s_3) \end{cases} \tag{8}$$

where τ_b and s_b are the peak bond stress (bond strength) and its corresponding slip; α and p are parameters that can be determined from curve fitting of experimental results.

5.2. CMR model

To overcome the drawback of the Malvar's model, the CMR model proposed by Cosenza et al. [51] was used to better represent the ascending branch of bond-slip curve for FRP-concrete bond:

$$\frac{\tau}{\tau_b} = \left(1 - exp\left(-\frac{s}{s_r}\right)\right)^{\beta} \quad (0 \leq s \leq s_b) \tag{9}$$

where s_r and β are parameters that are derived from curve fitting of experimental data. Since the CMR model is only for the ascending part, it is worth noting that it may be applicable to the bond at serviceability state level, while not capable of describing the complete bond behavior of structures till failure [56].

Fig. 16 displays the bond-slip relationship at the loaded end using the respective mBPE model and CMR model, of which the parameters of these models were determined from the curve fittings of the test results and summarized in Table 6. Since the descending branch of the mBPE model uses a linear expression to describe the post-peak bond behavior, the deviations were legitimately larger for those curves having larger fluctuations, as shown in the green dash dot lines in Fig. 16. The coefficient of determination (R^2) were 0.8864, 0.8090, 0.7706 and 0.6675 for the respective control specimen and conditioned specimens, indicating rough accuracy of the predictions. On the contrary, all the results predicted by the ascending branch of the mBPE model and CMR model matched well with the test results, as shown in the red solid lines and blue short dash dot lines in Fig. 16. Furthermore, the ascending branches of the bond-slip curves predicted by the CMR model performed better than those predicted by the mBPE model. The R^2 of the CMR model for all the specimens were greater than 0.98, yielding good accuracy.

For the unconditioned specimen, the calibrated parameters α and p, s_r and β were 0.6136, 0.0664, 0.5777 and 1.2459 for the double branch of the mBPE model and CMR model, respectively. Consider the most adverse case, the coupled FT cycles and AS solution, these calibrated parameters were changed to 0.4150, 0.0900, 0.7318 and 0.7219 for the mBPE model and CMR model, respectively. It is worth noting that these calibrated parameters determined from the experimental data were also dependent on the bond strength and its corresponding slip. The GFRP bars used in this study only consider the surface treatment of sand coated and helically wrapped. For different types of surface conditions, the calibrated parameters may vary accordingly and should be determined additionally.

Table 7 shows the calibrated average values for both mBPE model and CMR model considering different environmental conditions, and the values for unconditioned specimens are also listed. Based on the test results, α and p of mBPE model were suggested to be 0.6211 and 0.0675 for unconditioned scenario, 0.4166 and 0.0879 for FT cycles, 0.6009 and 0.0846 for AS solution, 0.4064 and 0.0897 for coupled FT cycles and AS solution. Also, s_r and β of CMR model were recommended to be 0.0613 and 1.2438 for unconditioned scenario, 0.7642 and 0.6800 for FT cycles, 0.6005 and 1.1722 for AS solution, 0.7365 and 0.7266 for coupled environment. The most detrimental environment attack on the bond behavior was encountered when the GFRP-concrete specimens were subjected to the coupled environmental conditions and thus, safe design values for the calibrated parameters are recommended as the values chosen from the coupled scenario.

6. Conclusions

This study presented a detailed experimental-analytical investigation on the bond durability of GFRP bars in concrete when exposed to simulated weathering. A total of 26 pullout samples with three different concrete covers were tested under different environmental conditions in terms of FT cycles, AS solution and the coupling of each. The durability of the GFRP-concrete specimens

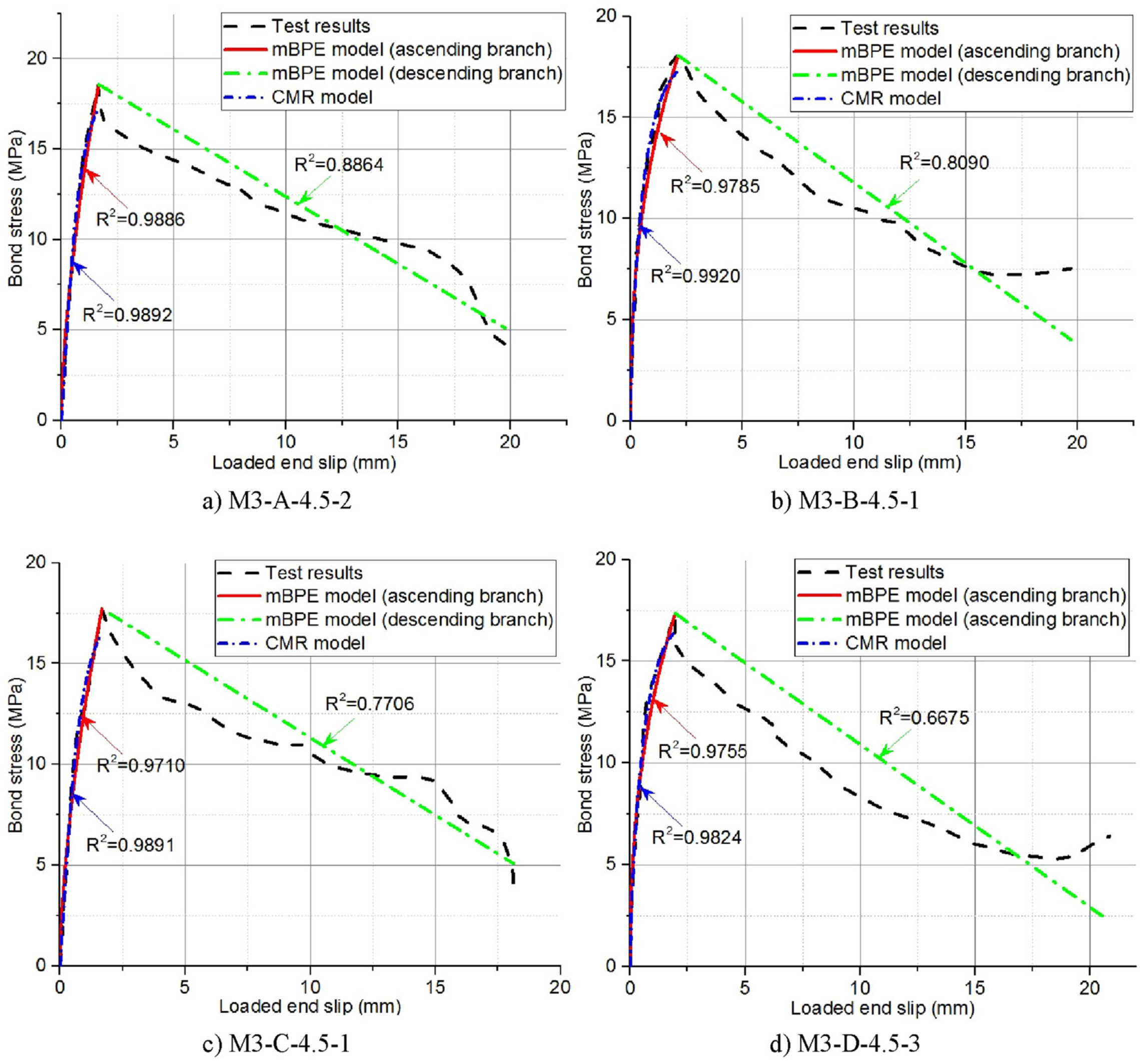


Fig. 16. Curve fittings of the typical bond stress-slip relationship.

Table 6
Fitting parameters of the mBPE model and CMR model.

Specimen	τ_b (MPa)	s_b (mm)	mBPE model		CMR model	
			α	p	s_r	β
M3-A-4.5-2	18.8898	1.4070	0.6136	0.0664	0.5777	1.2459
M3-B-4.5-1	18.0655	1.7512	0.4011	0.0940	0.7630	0.6768
M3-C-4.5-1	18.7163	1.6730	0.6087	0.0725	0.5911	1.1783
M3-D-4.5-3	17.3500	1.6537	0.4150	0.0900	0.7318	0.7219

Table 7
Mean values of the parameters considering different environmental conditions.

Environments	τ_b (MPa)	s_b (mm)	mBPE model		CMR model	
			α	p	s_r	β
Unconditioned	18.86	1.42	0.6211	0.0675	0.6013	1.2438
FT cycles	18.27	1.72	0.4166	0.0879	0.7642	0.6800
AS solution	18.58	1.67	0.6009	0.0846	0.6005	1.1722
Coupled FT cycles and AS solution	17.34	1.78	0.4064	0.0897	0.7365	0.7226

under weathering were assessed through the failure mode, weight loss, relative dynamic modulus of elasticity and durability factor, as well as the bond strength reduction. With the obtained experimental data, the analytical models were then calibrated by considering environmental influences for more widespread applications. Specifically, several conclusions can be drawn in the following:

(1) The failure mode of the pullout specimens having concrete cover $c = 3.0d_b$ changed from pullout of the bars to concrete splitting when exposed to 75 FT cycles with temperatures ranging from -18 ± 2 °C and 4 ± 2 °C, while pullout failure

was observed in all the specimens having concrete cover $c = 4.5d_b$. Such observations were consistent with the stipulations of ACI 440.1R-06, of which the bond equation accounts for pullout failure by limiting c/d_b to 3.5. From the perspective of design, concrete cover with sufficient resistance to prevent brittle failure of concrete splitting needs to be addressed especially for the weathering, such as FT effects.

(2) The surface scaling of the specimens subjected to the coupled FT cycles and AS solution was obvious, where the flaking of the surface mortar and exposure of the coarse aggregate were clearly observed. Also, the weight loss of those coupled conditioned specimens was the largest among all scenarios, with 1.23% and 1.34% for the specimens having $c = 3.0d_b$ and $c = 4.5d_b$, respectively. On the contrary, the specimens conditioned with AS solution exhibited minor weight losses, with 0.24% and 0.21% for the specimens having $c = 3.0d_b$ and $c = 4.5d_b$, respectively.

(3) The specimens having $c = 3.0d_b$ suffered from approximately 26% reduction in the relative dynamic modulus of elasticity under the coupled FT cycles and AS solution. Accordingly, the durability factors of the pullout specimens conditioned with coupled environments were generally smaller than those conditioned with individual FT cycles. The smallest durability factor was 51.21%, which was observed in the specimen with $c = 3.0d_b$ under the coupled environment. Furthermore, the durability factors of all the FT conditioned specimens fell into the interval between 40% and 85%, indicating the medium FT resistance.

(4) The bond-slip relationship reveals that increasing concrete cover may have great contribution in enhancing the pre-peak bond resistance to the environmental attacks in terms of FT cycles, AS solution, and coupled influences of the both. By increasing the concrete cover from $c = 3.0d_b$ to $c = 4.5d_b$, the bond strength reductions were reduced from 14% to 8% under the coupled conditioning, and from 12% to 3% under the individual FT cycles. Similar patterns were also observed in the normalized bond strength, indicating that larger concrete cover can effectively improve the FT resistance of the GFRP-concrete element. In addition, the individual AS solution was found to have minor impact on the bond strength of the pullout specimens.

(5) The calibrated analytical models considering the environmental effect matched well with the experimental results for the bond-slip prediction in terms of the ascending branch. In particular, the CMR model performed better than the mBPE model. The R^2 of the CMR model for all bond-slip curve-fittings were greater than 0.98, indicating rather close predictions to the test results. The average values of the unconditioned and conditioned specimens were summarized. Considering the worst case, the coupled conditioning, α and p of mBPE model were suggested to be 0.4064 and 0.0897, and s_r and β of CMR model were recommended to be 0.7365 and 0.7266, respectively.

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